ARASH JALIL KHABBAZI

arashjkh@purdue.edu
West Lafayette, IN, USA
ajkhabbazi.github.io
linkedin.com/in/ajkhabbazi
Google Scholar

SUMMARY

I'm a PhD candidate in Mechanical Engineering at Purdue University, working on modeling, control, optimization, and machine learning for building energy systems. My research builds practical automation for efficient, occupant- and grid-interactive buildings, focusing on predictive control and reinforcement learning. I develop and demonstrate scalable data-driven control algorithms in real buildings, and I also work on power control systems in homes and on demand flexibility in commercial buildings. My work combines rigorous methods with real-world testing to make control systems more reliable and easier to deploy at scale.

EDUCATION

Purdue University, *West Lafayette, IN* — *08/2023 – Present*
PhD in Mechanical Engineering — **GPA:** 3.9/4.0
Concentration in Computational Science and Engineering

University of British Columbia, *BC, Canada* — *05/2021 – 08/2023*
MS in Mechanical Engineering — **GPA:** 94% (4.0/4.0)
Thesis: "Mixing Gaseous Hydrogen into Natural Gas Distribution Pipelines"

University of Tabriz, *EA, Iran* — *08/2016 – 07/2020*
BS in Mechanical Engineering — **GPA:** 19.12/20 (4.0/4.0) – **Rank:** 1/155+

SELECT AWARDS AND RECOGNITION

ASHRAE Graduate Student Grant-in-Aid Award — *2025*
Best PhD Forum Presentation Runner-up Award, *ACM BuildSys* — *2025*
ACM SIGEnergy Student Travel Grant, *ACM BuildSys* — *2025*
Best Paper Award, *CSME International Congress* — *2023*
Best Presentation Award, *CSME International Congress (Advanced Energy Symposium)* — *2022*
Graduate Research Scholarship, *University of British Columbia* — *2022*
Graduate Dean's Entrance Scholarship, *University of British Columbia* — *2021*

PUBLICATIONS

Journal Articles

2. **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, P. Papageorgiou, A.H. Lee, J.E. Braun, G.P. Henze, and K.J. Kircher, "Lessons learned from field demonstrations of model predictive control and reinforcement learning for residential and commercial HVAC: A review." *Applied Energy, 2025.*
1. **A.J. Khabbazi**, M. Zabihi, R. Li, M. Hill, V. Chou, and J. Quinn, "Mixing hydrogen into natural gas distribution pipeline system through Tee junctions." *International Journal of Hydrogen Energy, 2024.*

Preprints

1. L.D. Reyes Premer, **A.J. Khabbazi**, and K.J. Kircher, "Indirect Data-Driven Predictive Control and the State-Space Predictor." *arXiv:2602.10936, 2026. Under review.*

Conference Papers

6. W.G. Dierking, **A.J. Khabbazi**, L.D. Reyes Premer, and K.J. Kircher, "Scalable Supervisory HVAC Control for Linear Objectives." *65th IEEE Conference on Decision and Control (CDC), 2026.*
5. L.D. Reyes Premer, E.N. Pergantis, **A.J. Khabbazi**, F. Wu, and K.J. Kircher, "Evaluating Power Control Strategies for Avoiding Residential Panel Upgrades in Cold Climates." *ACEEE Summer Study on Energy Efficiency in Buildings, 2026.*
4. **A.J. Khabbazi** and K.J. Kircher, "Small HVAC Control Demonstrations in Larger Buildings Often Overestimate Savings." *American Control Conference (ACC), 2026.*
3. E.N. Pergantis, L.D. Reyes Premer, **A.J. Khabbazi**, Priyadarshan, F. Wu, D. Ziviani, and K.J. Kircher, "Active current limiting control of residential appliances for breaker panel protection across the US: a preliminary study." *CISBAT, 2025.*

2. **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, A.H. Lee, J. Ma, H. Liu, G.P. Henze, and K.J. Kircher, "What Have We Learned From Field Demonstrations of Advanced Commercial HVAC Control?" *8th International High Performance Buildings Conference, 2024.*
1. **A.J. Khabbazi**, M. Zabihi, R. Li, V. Chou, and J. Quinn, "Blending of Hydrogen into a Natural Gas Distribution Pipeline in British Columbia through a Tee Junction for Reducing GHG Emissions." *Canadian Society for Mechanical Engineering (CSME) International Congress, 2023.* ***Best Paper Award.***

Abstracts

3. **A.J. Khabbazi**, "PhD Forum Abstract: Field Demonstration of Advanced Commercial HVAC Control for Scalable Deployment." *12th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys), 2025.* ***Best PhD Forum Presentation Runner-up Award.***
2. **A.J. Khabbazi**, R. Li, and J. Quinn, "Green Hydrogen Supply to Urban Infrastructure and Buildings through Blending into the Existing Grid." *Canadian Society for Mechanical Engineering (CSME) International Congress, 2022.* ***Best Presentation Award.***
1. **A.J. Khabbazi**, R. Li, J. Quinn, and M. Hoorfar, "The Blending and Transmission of Hydrogen and Natural Gas in Transmission and Distribution Pipelines." *13th International Green Energy Conference (IGEC-XIII), 2021.*

TALKS

8. "Small HVAC Control Demonstrations in Larger Buildings Often Overestimate Savings." *American Control Conference (ACC), New Orleans, LA, 2026.*
7. "Field Demonstration of Advanced Commercial HVAC Control for Scalable Deployment." *PhD Forum, 12th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys), Golden, CO, 2025.*
6. "What Have We Learned From Field Demonstrations of Advanced Commercial HVAC Control?" *8th International High Performance Buildings Conference, Purdue University, West Lafayette, IN, 2024.*
5. "Field Demonstrations of Advanced Commercial HVAC Control." *Intelligent Building Operations Workshop, Purdue University, West Lafayette, IN, 2024.*
4. "Mixing Gaseous Hydrogen into Natural Gas Distribution Pipelines." *Herrick Graduate Seminar, Purdue University, West Lafayette, IN, 2023.*
3. "Blending of Hydrogen into a Natural Gas Distribution Pipeline in British Columbia through a Tee Junction for Reducing GHG Emissions." *CSME International Congress, Université de Sherbrooke, QC, Canada, 2023.*
2. "Green Hydrogen Supply to Urban Infrastructure and Buildings through Blending into the Existing Grid." *CSME International Congress, University of Alberta, AB, Canada, 2022.*
1. "The Blending and Transmission of Hydrogen and Natural Gas in Transmission and Distribution Pipelines." *13th International Green Energy Conference (IGEC-XIII), 2021.*

RESEARCH EXPERIENCE

Purdue University, Graduate Research Assistant, Kircher Research Group, *West Lafayette, IN* *08/2023–Present*

- Building supervisory control software for a live field demonstration of data-driven HVAC control.
- Co-developed a commissioning-free LP-based supervisory HVAC controller, field-deployed in a home (CDC'26).
- Co-developed indirect data-driven predictive control (TPC) on a state-space predictor (under review).
- Derived a multi-zone model of how cross-zone heat transfer inflates measured HVAC savings (ACC'26).
- Co-developed current-limiting control for home appliances to avoid panel upgrades (CISBAT'25, ACEEE'26).
- Reviewed 100+ field demonstrations of MPC and RL for HVAC in real buildings (Applied Energy 2025).

INDUSTRY EXPERIENCE

Purdue Energy & Utilities, Graduate Researcher, Energy Systems, *West Lafayette, IN* *02/2026–Present*

- Leading modeling and pipelines for a pre-cooling pilot to shift demand off-peak on a 40,320-ton CHW system.
- Quantified ∼\$41k/yr savings and a statistically significant 8% lower approach from an air/dirt-separator retrofit.
- Delivered a nine-year EUI assessment of 21 renovations to Purdue leadership for the Giant Leaps Master Plan.

Tesla, Inc., Applied Machine Learning Intern, Drive Unit, *Sparks, NV* *05/2025–08/2025*

- Built MySQL pipelines and features; trained ML models (XGBoost, random forests, neural nets) with R^2 0.85.
- Helped deploy a thermal imaging system for drive-unit inspection, replacing destructive laboratory testing.
- Delivered ML quality gains projected to cut testing costs by \$1M+/year; offered a three-month extension.

FortisBC, Graduate Researcher, Renewable Gas, *Vancouver, BC* *05/2021–08/2023*

- Investigated hydrogen blending into natural gas distribution systems on an NSERC-funded UBC-FortisBC project.

- Performed CFD simulations and developed ideal- and real-gas mixture models in C.
- Contributed to hydrogen research within the UBC H_2Lab, established through this collaboration.
- Delivered research resulting in one journal paper, three conference papers, an MS thesis, and two awards.

TEACHING EXPERIENCE

Purdue University, Graduate Teaching Assistant (PhD)

- ME597: Distributed Energy Resources (DERs) — *Spring'26*
  - Supported 52 students on modeling and control of DERs such as batteries, solar PV, EVs, and HVAC.
  - Assisted with homework, coding modules, and office hours covering DER concepts.
  - Guided semester projects on DER topics, including building controls and data center cooling.

University of British Columbia, Graduate Teaching Assistant (MS)

- Engineering Analysis I — *Fall'21, Spring'22, Fall'22*
- Fluid Mechanics II Lab — *Spring'22, Spring'23*

LEADERSHIP & MENTORSHIP

Research Mentor, Purdue University — *2024–Present*

- Mentored three master's students and one undergraduate on research planning and technical writing.
- Collaborated on their research and co-authored the resulting publications.

Disability Resource Centre Invigilator, University of British Columbia — *2021–2023*

- Invigilated accommodated exams, supporting students with approved testing accommodations.

ACADEMIC SERVICE

Conference and workshop organizing:

July 2026	Student Chair, Intelligent Building Operations (IBO) Workshop, *West Lafayette, IN*
July 2024	Student Chair, Intelligent Building Operations (IBO) Workshop, *West Lafayette, IN*

Society memberships: ASHRAE, ACM, IEEE, CSME

Reviewing: IEEE Transactions on Control Systems Technology; IEEE Conference on Decision and Control; Building and Environment; Journal of Building Performance Simulation; ACEEE Summer Study on Energy Efficiency in Buildings; Herrick Conferences

SELECT COURSES

- ECE570: Artificial Intelligence
- IE690: Reinforcement Learning & Control (RLC)
- AAE568: Applied Optimal Control & Estimation
- AAE 561: Intro to Convex Optimization
- STAT511: Statistical Methods
- ME597: Industrial Internet of Things (IIoT)
- ME597: Distributed Energy Resources (DERs)
- ME511: Analysis of Thermal Systems

SKILLS

Languages:	Python, MATLAB, C, SQL
Optimization & ML:	CVXPY, Gurobi, TensorFlow, PyTorch, scikit-learn, XGBoost, Stable-Baselines3
Simulation & Testbeds:	BOPTEST, Gymnasium, OCHRE
Systems & Tools:	InfluxDB, Grafana, Git, Linux, LaTeX, Docker